Abhijit Marar

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Education

University of Texas at Dallas — **Aug 2024 – Present**
Doctor of Philosophy in Management Science — *Richardson, Texas*
Concentration: Operations Management

University of Georgia — **Aug 2015 – Dec 2020**
Doctor of Philosophy in Electrical Engineering — *Athens, Georgia*

Ruprecht-Karls Universität Heidelberg — **Sep 2013 – Aug 2015**
Master of Science in Biomedical Engineering — *Heidelberg, Germany*

University of Illinois at Urbana-Champaign — **Aug 2009 – Jun 2015**
Bachelor of Science in Electrical Engineering — *Champaign, Illinois*

Publications

1. **"Fundamental precision bounds for three-dimensional optical localization microscopy using self-interference digital holography"**, (with Peter Kner), 2021. ***Biomedical Optics Express***, 12(1), 20-40.

2. **"Roadmap on Recent Progress in FINCH Technology"**, (with Joseph Rosen, Simon Alford, Vijaykumar Anand, Jonathan Art, Petr Bouchal, Zdeněk Bouchal, Munkh-Uchral Erdenebat, Lingling Huang, Ayumi Ishii, Saulius Juodkazis, Nam Kim, Peter Kner, Takako Koujin, Yuichi Kozawa, Dong Liang, Jun Liu, Christopher Mann, Atsushi Matsuda, Teruyoshi Nobukawa, Takanori Nomura, Ryutaro Oi, Mariana Potcoava, Tatsuki Tahara, Bang Le Thanh, and Hongqiang Zhou), 2021. ***Journal of Imaging***, 7(10), 197.

3. **"Cell-type–specific, multicolor labeling of endogenous proteins with split fluorescent protein tags in Drosophila"**, (with Rie Kamiyama, Kota Banzai, Peiwei Liu, Ryo Tamura, Fangchao Jiang, Miyuki Fitch, Jin Xie, and Daichi Kamiyama), 2021. ***Proceedings of the National Academy of Sciences***, 118(23), e2024690118.

4. **"Imaging of In Vitro and In Vivo Neurons in Drosophila Using Stochastic Optical Reconstruction Microscopy"**, (with Melissa Ana Inal, Kathy Clara Bui, Shaoheng Li, Peter Kner, and Daichi Kamiyama), 2021. ***Current protocols***, 1(7), e203.

5. **"Three-dimensional nanoscale localization of point-like objects using self-interference digital holography"**, (with Peter Kner), 2020. ***Optics Letters***, 45(2), 591-594.

6. **"Charge-directed immobilization of bacteriophage on nanostructured electrode for whole-cell electrochemical biosensors"**, (with Yan Zhou, Peter Kner, and Ramaraja Ramasamy), 2017. ***Analytical chemistry***, 89(11), 5734-5741.

Work in Progress

1. **"Designing Admissions Requirements: Balancing Access, Competition, and Student Outcomes"**, (with Ignacio Rios)

Honors & Awards

Graduate School Student Travel Award	2019 & 2020
Innovative and Interdisciplinary Research Grant	2017
Five-year Ph.D. Fellowship, UT Dallas Jindal School of Management	2024 – 2029
Five-year Ph.D. Fellowship, UGA College of Engineering	2015 – 2020

Professional Experience

AVR Optics **May 2023 – Jul 2024**
Product Specialist - Confocal Microscopy & Spatial Light Modulators *Fairport, New York*

- Product manager responsible for the launch and development of a new confocal microscopy product line and the spatial light modulator product line, overseeing all aspects of market research, product positioning, customer engagement, and sales strategy to drive its successful establishment and growth within the market.
- Generated a robust and consistent sales pipeline, securing 2M USD in potential revenue within the initial five months in the role.

Excelitas Technologies Inc. **Jun 2022 – Jan 2023**
Business Development Manager *Waltham, Massachusetts*

- Sales manager responsible for business development in the Northeastern USA ($\sim$ 9M USD sales/yr).
- Achieved a $\sim$67% increase in territorial revenue within six months of hire by successfully leading negotiations for long-term contracts with three existing customers.
- Developed and presented a strategic plan to enter a new market (quantum technologies, $\sim$ 4.5B USD market size) resulting in new product development roadmap and M&A strategy.

TOPTICA Photonics **Oct 2021 – Jun 2022**
Scientific Sales - Biophotonics and Materials Applications *Rochester, New York*

- Sales manager responsible for biophotonics and material applications covering the North American territory ($\sim$ 3M USD sales/yr) Responsible for development and growth of ultrafast product line ($\sim$ 2M USD sales/yr)
- Secured six strategically important >150K USD sales of terahertz systems exceeding sales targets by 50% for terahertz systems within two quarters.

Georgia Institute of Technology **Mar 2021 – Sep 2021**
Postdoctoral Fellow *Atlanta, Georgia*

- Developed the optics, mechanics and software for a hyperspectral light field microscope that allows single shot hyperspectral volumetric imaging (five channels - XYZTλ) of fluorescently labelled biological samples.

Conference Proceedings

1. **"Three-dimensional localization precision for self-interference digital holograph"**, (with Peter Kner), 2021. ***SPIE Photonics West***, 116500P.

2. **"Three-dimensional super-resolution imaging with self-interference digital holography"**, (with Peter Kner), 2021. ***Digital Holography and Three-Dimensional Imaging***, JTh4D-4.

3. **"Self-interference digital holography (SIDH) for far-field three-dimensional nanoscale localization of fluorescent nanoparticles"**, (with Peter Kner), 2020. ***SPIE Photonics West***, 112460M.

4. **"Three-dimensional localization microscopy by incoherent holography"**, (with Peter Kner), 2019. ***SPIE Photonics West***, 108840I.

5. **"Synthesis of Photoswitchable Quantum Dots for Superresolution Microscopy"**, (with Abhilasha Dehankar, Kil Ho Lee, Thomas Porter, Karine Thate, Carol Lynn Alpert Peter Kner, and Jessica Winter), 2018. ***AIChE Annual Meeting***, 196j.

6. **"Photoswitchable Quantum Dots Probes for Superresolution Microscopy"**, (with Abhilasha Dehankar, Kil Ho Lee, Karine Thate, Carol Lynn Alpert, Peter Kner, and Jessica Winter), 2017. ***AIChE Annual Meeting***, 199c.

7. **"Measuring the Change of Mitochondrial Morphology in RSV Infected MHS Cells using STORM"**, (with Kenya Alfaro, Jocelyn Grunwell, and Peter Kner), 2017. ***BMES Annual Meeting***.

8. **"Wavefront correction for superresolution microscopy"**, (with Peter Kner, Kayvan Tehrani, and Benjamin Thomas), 2017. ***SPIE Nanoscience + Engineering***, 1035006.

9. **"SLIM: Tool to aid pathologists in separation of Gleason grade 3 and 4"**, (with Shamira Sridharan, Jon Liang, Le Wang, Karan Shah, Ryan Tapping, Virgilia Macias, Krishnarao Tangella, Andre Kajdacsy-Balla, and Gabriel Popescu), 2013. ***United States and Canadian Academy of Pathology's 102nd Annual Meeting***.

Teaching Experience

Marine Biological Laboratory **2018 – 2019**
Teaching Assistant, Analytical and Quantitative Light Microscopy *Woods Hole, Massachusetts*

- Designed and delivered lectures while overseeing laboratory sessions focusing on advanced optical microscopy techniques for a diverse audience, encompassing 39 graduate students, post-doctoral researchers, and faculty.

Other Skills

Programming: Python, MATLAB, C++, CUDA, LaTeX, R, STATA
Languages: English (native), Hindi (fluent)

References

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